Factoring with special products

1

a $p+8$

b $m+6$

2 $x-7$

3 $\left(x^2-10\right)\left(x^2+10\right)$

4 $(ax-b)(ax+b)$

5

a $(x+5)(x-5)$

b $(n-6)(n+6)$

c $(v-7)(v+7)$

d $\left(x-\frac{2}{7}\right)\left(x+\frac{2}{7}\right)$

e $(9-x)(9+x)$

f $(2-u)(2+u)$

g $(2+7y)(2-7y)$

h $(8x-11)(8x+11)$

i $(11m+8)(11m-8)$

j $(9x-10y)(9x+10y)$

k $(11x+7y)(11x-7y)$

l $(4x-5)(4x+5)$

m $6(x-7)(x+7)$

n $3(t+2)(t-2)$

o $5(x-8)(x+8)$

p $4(n+2)(n-2)$

6

a $(x-7)(x+7)$ square units

b $(4x-3)(4x+3)$ square units

7 140

8 388

9 $39\,999$

10

a $(a+b)^2$

b

i a^2

ii ab

iii ab

iv b^2

c $a^2+2ab+b^2$

d Perfect square identity.

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11 **a** $(b+7)^2$ **b** $(y-5)^2$ **c** $(x+5)^2$ **d** $(x+7)^2$

e $(x-6)^2$ **f** $(x-8)^2$ **g** $(9x-4)^2$ **h** $(9x+2)^2$

i $(7x-2)^2$ **j** $(9+x)^2$ **k** $(6-x)^2$ **l** $(5+6r)^2$

m $7(d+8)^2$ **n** $4(x+5)^2$ **o** $3(x-4y)^2$ **p** $-6(r+8)^2$

q $-3(x-2)^2$ **r** $\left(b-\frac{7}{2}\right)^2$ **s** $\left(x+\frac{3}{2}\right)^2$ **t** $\left(x-\frac{3}{2}\right)^2$

12

a **i** $(x+6)^2$ **ii** $x+6$

b **i** $3(2x+1)^2$ **ii** $2x+1$

c **i** $6(x+3)^2$ **ii** $x+3$

13 9

Additional questions

14

a $9x^2-16$ **b** $3x+4$ and $3x-4$

15 Answers may vary.
$9x^2+54x+81$ and $4x^2+36x+81$ are two possibilities.

16

$$\begin{aligned}(a^2-b^2)(c^2-d^2) &= a^2c^2-a^2d^2-b^2c^2+b^2d^2\\ &= a^2c^2+b^2d^2-a^2d^2-b^2c^2\\ &= a^2c^2+2abcd+b^2d^2-a^2d^2-2abcd-b^2c^2\\ &= a^2c^2+2abcd+b^2d^2-\left(a^2d^2+2abcd+b^2c^2\right)\\ &= (ac+bd)^2-(ad+bc)^2\end{aligned}$$

17 Gabriella did not pull out the common factor correctly. Since there is a factor of 2 in *both* factors of her expression, she needs to factor out a greatest common factor of 4.

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a The total area of the blanket is $9x^2 + 45x + 54$. Many factorizations exist. Three possible answers are: $(3x + 6)(3x + 9)$, $9(x + 2)(x + 3)$, or $9\left(x^2 + 5x + 6\right)$

b Answers will vary. Since dimensions can't be zero or negative, all answers must have $x > 2.5$.
For example, if $x = 10$ inches, the overall dimensions for the blanket would be 36 inches by 39 inches and each piece would have dimensions of 21 inches, 15 inches, 23 inches, 13 inches, 16 inches and 10 inches which will all reasonably display a photograph.

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